EXPERIMENT 16: Experiment to understand EDA-Quantitative and Qualitative analysis.

DESCRIPTION:

Understand the importance of EDA-Quantitative and Qualitative analysis.

OBJECTIVE:

The program performs quantitative and qualitative EDA using Pandas, NumPy, and Matplotlib. It analyzes numerical data using statistical measures and categorical data using frequency counts and visualizations.

PROGRAM:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

data = {

    "total_bill": [12.50, 18.75, 25.40, 32.60, 15.80, 27.90, 21.50, 35.20, 19.40, 14.60],

    "tip": [1.50, 2.25, 3.80, 4.20, 2.00, 3.50, 2.80, 5.00, 2.50, 1.80],

    "sex": ["Male", "Female", "Male", "Female", "Male", "Female", "Male", "Female", "Male", "Female"],

    "smoker": ["Yes", "No", "Yes", "No", "No", "Yes", "No", "Yes", "No", "Yes"],

    "day": ["Mon", "Mon", "Tue", "Tue", "Wed", "Wed", "Thu", "Thu", "Fri", "Fri"],

    "time": ["Lunch", "Dinner", "Lunch", "Dinner", "Lunch", "Dinner", "Lunch", "Dinner", "Lunch", "Dinner"],

    "size": [1, 2, 3, 4, 2, 5, 3, 6, 2, 4]

}

tips = pd.DataFrame(data)

print("Dataset:")

print(tips)

# Quantitative Analysis

print("\nQuantitative Analysis:")

print(tips.describe())

# Qualitative Analysis

print("\nQualitative Analysis:")

print(tips["time"].value_counts())
```

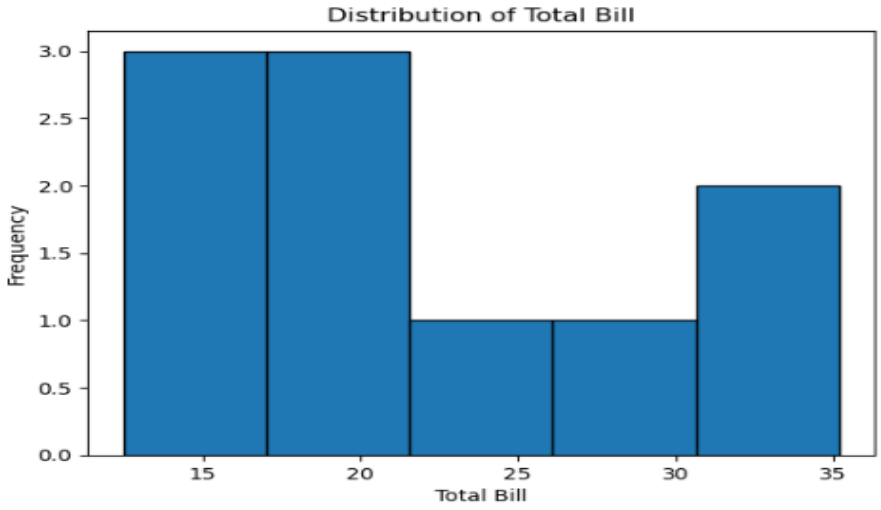
```
print(tips["day"].value_counts())
# Distribution of Total Bill
plt.hist(tips["total_bill"], bins=5, edgecolor="black")
plt.xlabel("Total Bill")
plt.ylabel("Frequency")
plt.title("Distribution of Total Bill")
plt.show()
# Tip vs Total Bill
plt.scatter(tips["tip"], tips["total_bill"])
plt.xlabel("Tip")
plt.ylabel("Total Bill")
plt.title("Tip vs Total Bill")
plt.show()
# Tip vs Total Bill with trend line
plt.scatter(tips["tip"], tips["total_bill"])
m, b = np.polyfit(tips["tip"], tips["total_bill"], 1)
plt.plot(tips["tip"], m * tips["tip"] + b)
plt.xlabel("Tip")
plt.ylabel("Total Bill")
plt.title("Tip vs Total Bill with Regression Line")
plt.show()
# Box plot
plt.boxplot(tips["total_bill"])
plt.ylabel("Total Bill")
plt.title("Box Plot of Total Bill")
plt.show()
# Bar chart for Time
tips["time"].value_counts().plot(kind="bar")
plt.xlabel("Time")
plt.ylabel("Count")
plt.title("Customers by Time")
plt.show()
```

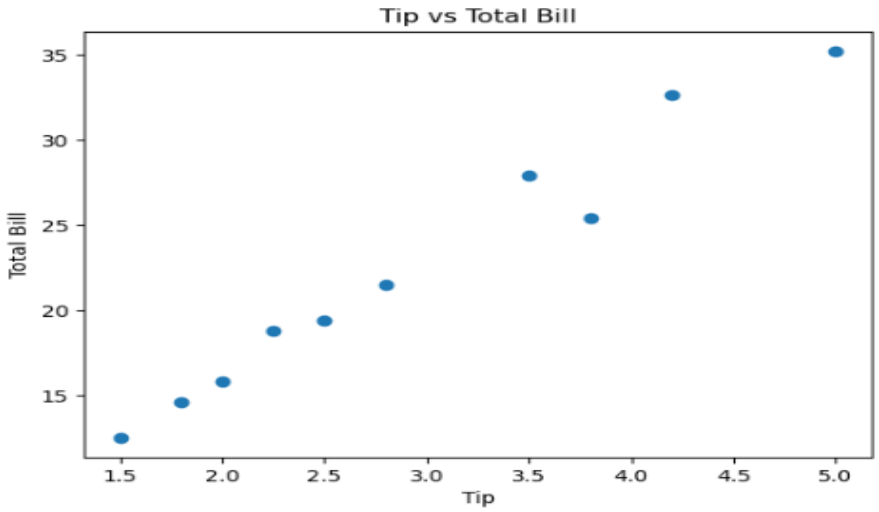
OUTPUT:

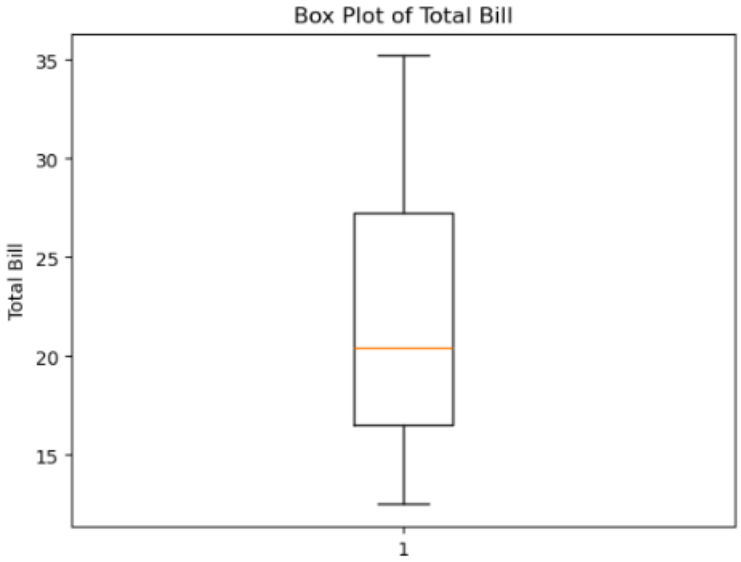
```
Dataset:
   total_bill   tip     sex smoker  day    time  size
0       12.50  1.50    Male    Yes  Mon   Lunch     1
1       18.75  2.25  Female     No  Mon  Dinner     2
2       25.40  3.80    Male    Yes  Tue   Lunch     3
3       32.60  4.20  Female     No  Tue  Dinner     4
4       15.80  2.00    Male     No  Wed   Lunch     2
5       27.90  3.50  Female    Yes  Wed  Dinner     5
6       21.50  2.80    Male     No  Thu   Lunch     3
7       35.20  5.00  Female    Yes  Thu  Dinner     6
8       19.40  2.50    Male     No  Fri   Lunch     2
9       14.60  1.80  Female    Yes  Fri  Dinner     4

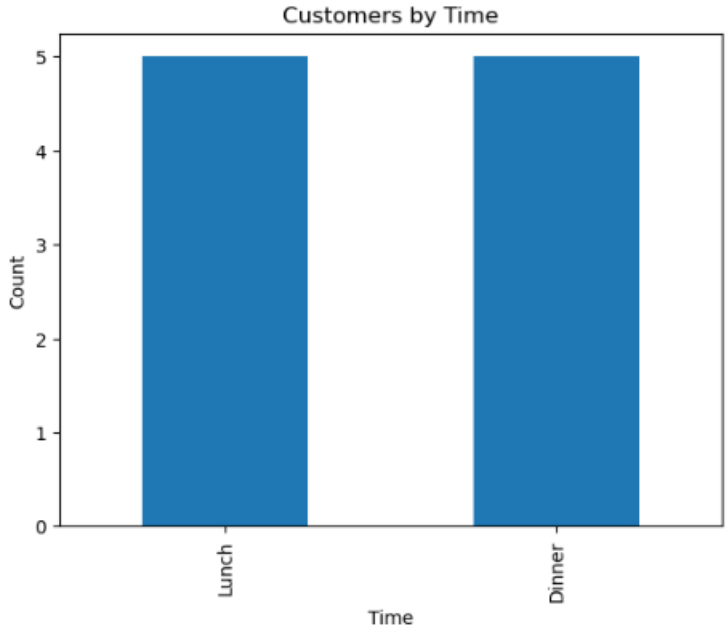
Quantitative Analysis:
       total_bill        tip       size
count   10.000000  10.000000  10.000000
mean    22.365000   2.935000   3.200000
std      7.693144   1.146504   1.549193
min     12.500000   1.500000   1.000000
25%     16.537500   2.062500   2.000000
50%     20.450000   2.650000   3.000000
75%     27.275000   3.725000   4.000000
max     35.200000   5.000000   6.000000

Qualitative Analysis:
time
Lunch     5
Dinner    5
Name: count, dtype: int64
day
Mon    2
Tue    2
Wed    2
Thu    2
Fri    2
Name: count, dtype: int64
```






Box Plot of Total Bill
Total Bill
35
30
25
20
15
1


Customers by Time
Count
5
4
3
2
1
0
Lunch
Dinner
Time